Ying Pei Lin

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Education

B.S. Mechanical Engineering, NYCU, Taiwan *2021-09 - 2026-06 (ongoing)*

- Coursework: Control Systems, Robotics, Microcontrollers.
- GPA: 3.75

Exchange Student, KTH Royal Institute of Technology, Sweden *2024-08 - 2025-06*

- Coursework: Machine Learning, Deep Learning in Data Science, Safe Robot Planning and Control.

Skills

Programming	Python • C/C++/C# • Java • MATLAB • JavaScript
Design	Unity • SolidWorks • Blender
Hardware	STM32 • Arduino • Raspberry Pi
Robotics / ML	ROS • FreeRTOS • SLAM • PyTorch • YOLO

Experience

Kibo Robot Programming Competition *2025-04 - 2026-02*

- **World Champion & 1st Place in Taiwan** – Designed modular robotic system for autonomous vision and navigation in the ISS simulator.
- Developed YOLOv11-based detection (99% precision) and revised weighted box fusion algorithm.
- Program executed aboard the ISS, representing Taiwan internationally.

Meichu Hackathon – Logitech Challenge *2025-10*

- Awarded **3rd Place** for an AR interior-design assistant on Meta Quest with Logitech MX Ink spatial sketching.
- Captured real-time 3D sketches and generated AI-driven 3D furniture prototypes.
- Built a Unity pipeline (Line → Mask → Image → Mesh) and deployed as an installable APK.

Asfaloth Rocket Launch *2025 Summer*

- Programmed a Raspberry Pi data acquisition system using the Prophesee OpenEB SDK to operate an event-based camera payload for a sounding rocket.

Intern at Industrial Technology Research Institute (ITRI) *2023-07 - 2023-12*

- Built interactive 3D LiDAR/GPS web visualizer (ROS + JavaScript) for mechatronics lab

TDK Robocon Competition *2022 & 2024*

- 2022: Arduino-based motion control & motor driver system
- 2024: Embedded SLAM navigation (Hector + AMCL) on resource-limited robot

Projects

Quadruped Robot *2023 - 2025*

- Implemented full kinematic framework (forward, inverse, differential, body) for a 3D-printed quadruped robot.
- Developed locomotion control and validated stability through PyBullet simulation.
- Applied PPO-based reinforcement learning for adaptive gait optimization.

Character-level Language Modeling *2025-04 - 2025-06*

- Implemented Byte Pair Encoding (BPE) with 2000-token vocabulary on Shakespeare text.
- Trained RNN and LSTM models using BPE; improved 3-gram match frequency to 67.5% (vs. 46.8% in character-level baseline).
- Analyzed BPE's impact on text coherence and model loss in small-data settings.

Minecraft Exploration Agent *2023-04 - 2023-06*

- Built a DQN-based exploration system in Minecraft, using CNN to process 9-block observations.
- Optimized Q-learning with rewards and penalties, increasing navigation success rate.

Your expectations for this course

希望藉由這堂課學習有關感測器、機器人與自駕車相關的知識，並藉由實做熟悉背後的理論。

Personal image